		Data Type xsd:string
D_337 Time	Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 6 **max:** 6 **External Category** Element **Data Type** xsd:string
D_28 Group Control Number	Assigned number originated and maintained by the sender	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 9 **max:** 9 **External Category** Element **Data Type** xsd:string
D_455 Responsible Agency Code	Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480 **Selected Values** All Values in Codelist	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 1 **max:** 1 **External Category** Element **Data Type** xsd:string **Codelist Id** D_455 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_480 Version / Release / Industry Identifier Code	Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed **Selected Values** All Values in Codelist	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 6 **max:** 6 **External Category** Element **Data Type** xsd:string **Codelist Id** D_480 **Type System** Customer_TS **Version Mode** Local **Version** 1.1

2.5 S_GE — Functional Group Trailer

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **FunctionalGroup** — Functional Group	1 .. 1				
│ └ **S_GE** — Functional Group Trailer	1 .. 1				
│ ├ **D_97** — Number of Transaction Sets Included	1 .. 1	String		1 .. 1	
│ └ **D_28** — Group Control Number	1 .. 1	String		9 .. 9	

Documentation

Definition To indicate the end of a functional group and to provide control information Comments 1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header. Semantic Notes: 1. The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

Properties

Identifier S_GE

Name Functional Group Trailer

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_97 Number of Transaction Sets Included	Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string
D_28 Group Control Number	Assigned number originated and maintained by the sender	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 9 max: 9 **External Category** Element **Data Type** xsd:string

2.6 S_IEA — Interchange Control Trailer

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
└ **S_IEA** — Interchange Control Trailer	1 .. 1				
├ **D_I16** — Number of Included Functional Groups	1 .. 1	String		1 .. 1	
└ **D_I12** — Interchange Control Number	1 .. 1	String		9 .. 9	

Documentation

Definition To define the end of an interchange of zero or more functional groups and interchange-related control segments

Properties

Identifier S_IEA

Name Interchange Control Trailer

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_I16 Number of Included Functional Groups	A count of the number of functional groups included in an interchange	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string
D_I12 Interchange Control Number	A control number assigned by the interchange sender	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 9 max: 9 **External Category** Element **Data Type** xsd:string

3 OVERVIEW MESSAGE

General Information

Name	SAP_LBN_ANSI_X12_997
Direction	Out
Status	Draft
Message Type	Functional Acknowledgment
Type System	ASC_X12
Version	004010
External Category	Transaction Set

Documentation

Summary 997 is a functional ACK.

Definition This Draft Standard for Trial Use contains the format and establishes the data contents of the Functional Acknowledgment Transaction Set (997) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to define the control structures for a set of acknowledgments to indicate the results of the syntactical analysis of the electronically encoded documents. The encoded documents are the transaction sets, which are grouped in functional groups, used in defining transactions for business data interchange. This standard does not cover the semantic meaning of the information encoded in the transaction sets.

Notes

Usage 1

These acknowledgments shall not be acknowledged, thereby preventing an endless cycle of acknowledgments of acknowledgments. Nor shall a Functional Acknowledgment be sent to report errors in a previous Functional Acknowledgment.
The Functional Group Header Segment (GS) is used to start the envelope for the Functional Acknowledgment Transaction Sets. In preparing the functional group of acknowledgments, the application sender's code and the application receiver's code, taken from the functional group being acknowledged, are exchanged; therefore, one acknowledgment functional group responds to only those functional groups from one application receiver's code to one application sender's code.
There is only one Functional Acknowledgment Transaction Set per acknowledged functional group.

Usage 2 The EDI 997 message is triggered by SAP Business Network for Logistics as a response to acknowledge receipt of a file that the Carrier sends out. Hence, in this case. the Interchange Sender ID in the Header segment is always LBN instead of the Shipper LBN ID.

Usage 3 AK1 is used to respond to the functional group header and to start the acknowledgement for a functional group. There shall be one AK1 segment for the functional group that is being acknowledged.

Usage 4 AK2 is used to start the acknowledgement of a transaction set within the received functional group. The AK2 segments shall appear in the same order as the transaction sets in the functional group that has been received and is being acknowledged.

Comment 1 The data segments of this standard are used to report the results of the syntactical analysis of the functional groups of transaction sets; they report the extent to which the syntax complies with the standards for transaction sets and functional groups. They do not report on the semantic meaning of the transaction sets (for example, on the ability of the receiver to comply with the request of the sender).

3.1 Structure

The following table shows the complete structure.

Node	Card.	Prim.Type	Pos.	Length	Codelists
997 — Functional Acknowledgment	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
│ ├ **143** — Transaction Set Identifier Code	1 .. 1	Token	01	3 .. 3	143
│ └ **329** — Transaction Set Control Number	1 .. 1	String	02	4 .. 9	
├ **AK1** — Functional Group Response Header	1 .. 1		020		
│ ├ **479** — Functional Identifier Code	1 .. 1	Token	01	2 .. 2	479
│ └ **28** — Group Control Number	1 .. 1	Integer	02	1 .. 9	
├ **AK9** — Functional Group Response Trailer	1 .. 1		070		
│ ├ **715** — Functional Group Acknowledge Code	1 .. 1	Token	01	1 .. 1	715
│ ├ **97** — Number of Transaction Sets Included	1 .. 1	Integer	02	1 .. 6	
│ ├ **123** — Number of Received Transaction Sets	1 .. 1	Integer	03	1 .. 6	
│ └ **2** — Number of Accepted Transaction Sets	1 .. 1	Integer	04	1 .. 6	
└ **SE** — Transaction Set Trailer	1 .. 1		080		
├ **96** — Number of Included Segments	1 .. 1	Integer	01	1 .. 10	
└ **329** — Transaction Set Control Number	1 .. 1	String	02	4 .. 9	

4 DETAILS MESSAGE

4.1 997 — Functional Acknowledgment

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
997 — Functional Acknowledgment	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
├ **AK1** — Functional Group Response Header	1 .. 1		020		
├ **AK9** — Functional Group Response Trailer	1 .. 1		070		
└ **SE** — Transaction Set Trailer	1 .. 1		080		

Properties

Identifier 997

Name Functional Acknowledgment

Cardinality min: 1 max: 1

4.2 ST — Transaction Set Header

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
997 — Functional Acknowledgment	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
│ ├ **143** — Transaction Set Identifier Code	1 .. 1	Token	01	3 .. 3	143
│ └ **329** — Transaction Set Control Number	1 .. 1	String	02	4 .. 9	

Documentation

Definition To indicate the start of a transaction set and to assign a control number

Properties

Identifier ST

Name Transaction Set Header

Cardinality min: 1 max: 1

Syntax Type Related

External Category Segment

Position 010

Level 2

Notes

Usage 1 The transaction set identifier (ST01) used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
143 Transaction Set Identifier Code	Code uniquely identifying a Transaction Set **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** Token **Length** min: 3 max: 3 **External Category** Simple Data Element **Data Type** ID **Position** 01 **Codelist Id** 143 **Type System** ASC_X12 **Version Mode** Current **Version** 004010
329 Transaction Set Control Number	Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 4 max: 9 **External Category** Simple Data Element **Data Type** AN **Position** 02

4.3 AK1 — Functional Group Response Header

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
997 — Functional Acknowledgment	1 .. 1				
├ **AK1** — Functional Group Response Header	1 .. 1		020		
│ ├ **479** — Functional Identifier Code	1 .. 1	Token	01	2 .. 2	479
│ └ **28** — Group Control Number	1 .. 1	Integer	02	1 .. 9	

Documentation

Definition To start acknowledgment of a functional group

Properties

Identifier AK1

Name Functional Group Response Header

Cardinality min: 1 max: 1

Syntax Type Related

External Category Segment

Position 020

Level 2

Notes

Usage 1 AK101 is the functional ID found in the GS segment (GS01) in the functional group being acknowledged.

Usage 2 AK102 is the functional group control number found in the GS segment in the functional group being acknowledged.

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
479 Functional Identifier Code	Code identifying a group of application related transaction sets **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** Token **Length** min: 2 max: 2 **External Category** Simple Data Element **Data Type** ID **Position** 01 **Codelist Id** 479 **Type System** ASC_X12 **Version Mode** Current **Version** 004010
28 Group Control Number	Assigned number originated and maintained by the sender	**Cardinality** min: 1 max: 1 **Primitive Type** Integer **Length** min: 1 max: 9 **Fraction Digits** 0 **Total Digits** 9 **External Category** Simple Data Element **Data Type** N0 **Position** 02

4.4 AK9 — Functional Group Response Trailer

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
997 — Functional Acknowledgment	1 .. 1				
├ **AK9** — Functional Group Response Trailer	1 .. 1		070		
│ ├ **715** — Functional Group Acknowledge Code	1 .. 1	Token	01	1 .. 1	715
│ ├ **97** — Number of Transaction Sets Included	1 .. 1	Integer	02	1 .. 6	
│ ├ **123** — Number of Received Transaction Sets	1 .. 1	Integer	03	1 .. 6	
│ └ **2** — Number of Accepted Transaction Sets	1 .. 1	Integer	04	1 .. 6	

Documentation

Definition To acknowledge acceptance or rejection of a functional group and report the number of included transaction sets from the original trailer, the accepted sets, and the received sets in this functional group

Properties

Identifier AK9

Name Functional Group Response Trailer

Cardinality min: 1 max: 1

Syntax Type Related

External Category Segment

Position 070

Level 2

Notes

Comment 1 If AK901 contains the value “A” or “E”, then the transmitted functional group is accepted.

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
715 Functional Group Acknowledge Code	Code indicating accept or reject condition based on the syntax editing of the functional group **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** Token **Length** min: 1 max: 1 **External Category** Simple Data Element **Data Type** ID **Position** 01 **Codelist Id** 715 **Type System** ASC_X12 **Version Mode** Current **Version** 004010
97 Number of Transaction Sets Included	Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element	**Cardinality** min: 1 max: 1 **Primitive Type** Integer **Length** min: 1 max: 6 **Fraction Digits** 0 **Total Digits** 6 **External Category** Simple Data Element **Data Type** N0 **Position** 02
123 Number of Received Transaction Sets	Number of Transaction Sets received	**Cardinality** min: 1 max: 1 **Primitive Type** Integer **Length** min: 1 max: 6 **Fraction Digits** 0 **Total Digits** 6 **External Category** Simple Data Element **Data Type** N0 **Position** 03

2 Number of Accepted Transaction Sets	Number of accepted Transaction Sets in a Functional Group	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** Integer **Length** **min:** 1 **max:** 6 **Fraction Digits** 0 **Total Digits** 6 **External Category** Simple Data Element **Data Type** N0 **Position** 04